

C Programming

2_loops

loops

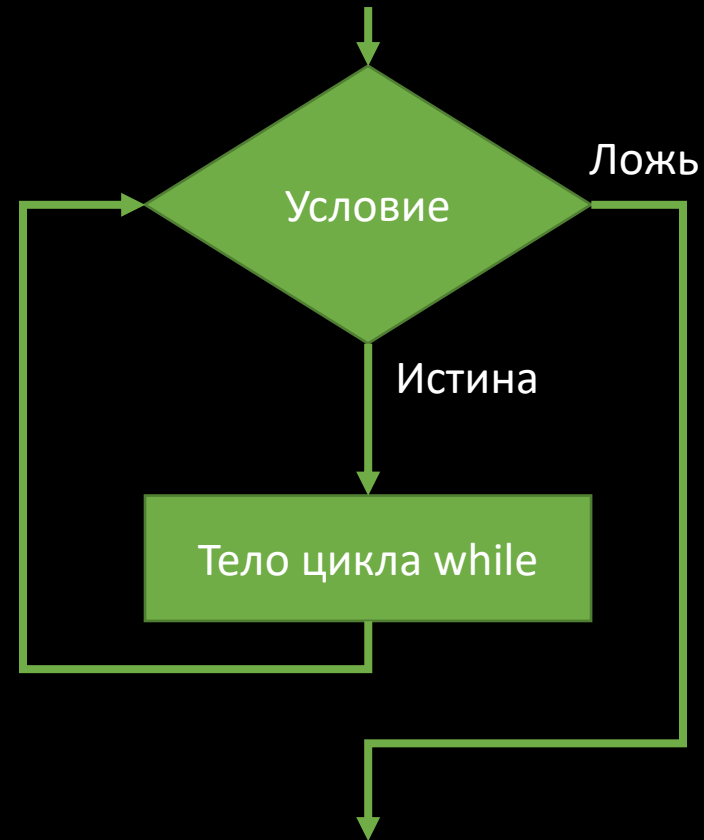


```
printf("meow\n");  
printf("meow\n");  
printf("meow\n");
```



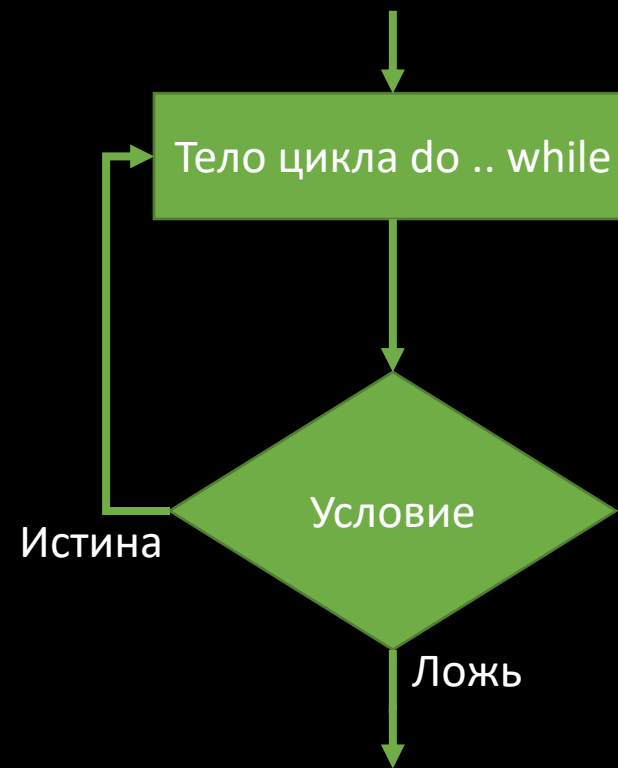
while loop (цикл с предусловием)

```
while (condition) {  
    // code block to be executed  
}
```



do .. while loop (цикл с постусловием)

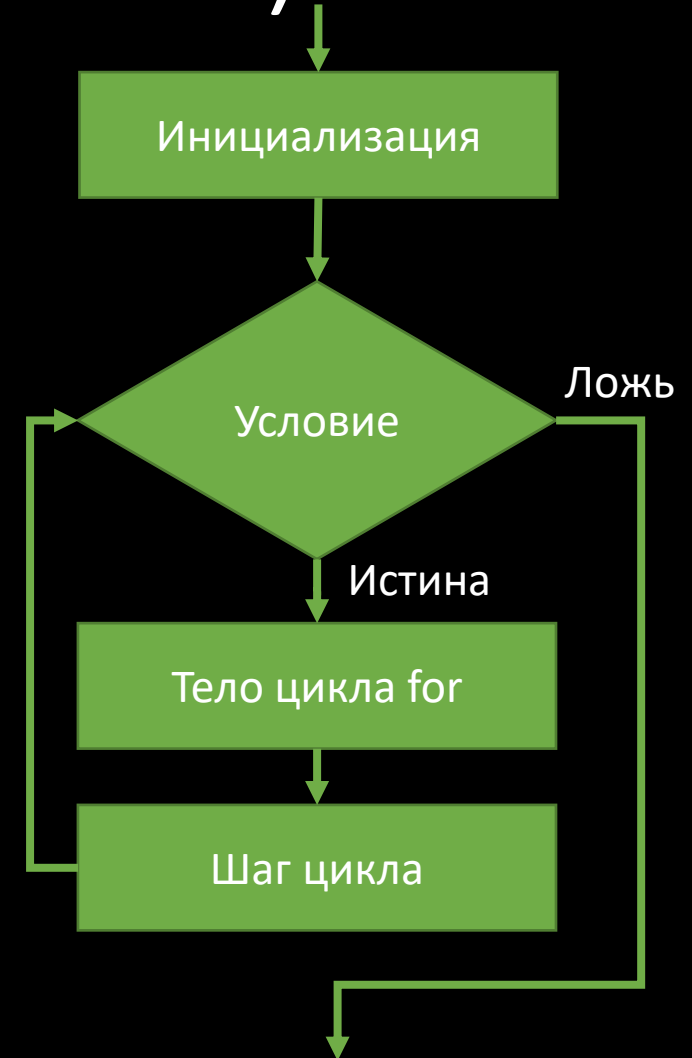
```
do {  
    // code block to be executed  
} while (condition);
```



for loop (цикл for)

```
for (s1; s2; s3) {  
    // code block to be executed  
}
```

- s1 – блок инициализации
- s2 – логическое выражение
- s3 – шаг цикла



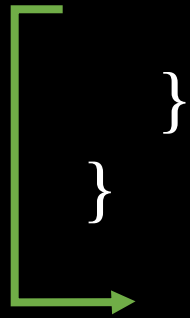
for loop variations

1. `for (s1; s2; s3) { ... }`
2. `for (; s2; s3) { ... }`
3. `for (s1;; s3) { ... }`
4. `for (s1; s2;) { ... }`
5. `for (; s2;) { ... }`
6. `for (s1;;) { ... }`
7. `for (;; s3) { ... }`
8. `for (;;) { ... }`

- s1 – блок инициализации
- s2 – логическое выражение
- s3 – шаг цикла

break

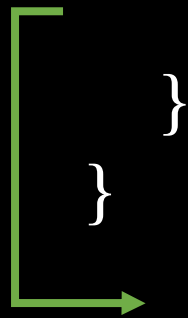
```
while (expr) {  
    if (expr_to_break) {  
        break;
```



```
do {  
    if (expr_to_break) {  
        break;
```



```
for (init; expr; step) {  
    if (expr_to_break) {  
        break;
```



continue

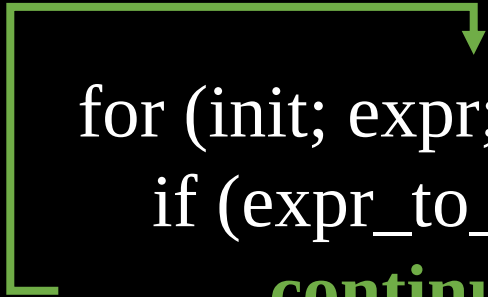
```
while (expr) {  
    if (expr_to_continue) {  
        continue;  
    }  
}
```



```
do {  
    if (expr_to_continue) {  
        continue;  
    }  
} while (expr);
```



```
for (init; expr; step) {  
    if (expr_to_continue) {  
        continue;  
    }  
}
```



Nested loops (Вложенные циклы)

```
for (int i = 1; i <= 3; i++) { // Внешний
    for (int j = 1; j <= 3; j++) { // Внутренний
        printf("%d ", i * j);
    }
    printf("\n");
}
```

Nested loops break/continue

```
for (s11; s12; s13) {  
    for (s21; s22; s23) {  
        continue;  
        // ...  
    }  
}  
  
for (s11; s12; s13) {  
    for (s21; s22; s23) {  
        // ...  
    }  
    break;  
}
```

Количество инструкций?

```
for (int i = 1; i <= 10; i++)  
{  
    a = a + i; // Считаем за одну  
}
```

```
for (int i = 1; i <= 10; i++)  
{  
    a = a + i; // Считаем за одну  
}
```

i = 1:	1
i <= 10:	11
i++:	10
a=a+i:	10
Всего:	32

Количество инструкций?

```
for (int i = 1; i <= 5; i++)  
{  
    a = a + i; // Считаем за одну  
    a = a + (5 + i); // Считаем за одну  
}
```

for loop unroll

```
for (int i = 1; i <= 10; i++)  
{  
    a = a + i;  
}
```

Количество инструкций
 $1 + 11 + 10 + 10 = 32$

```
for (int i = 1; i <= 5; i++)  
{  
    a = a + i;  
    a = a + (5 + i);  
}
```

Количество инструкций $1 + 6$
 $+ 5 + 10 = 22$

Github

<https://github.com/kruffka/C-Programming>

